Lecture 0

Prerequisites: A Refresher

This course assumes you can already *compute* with the tools of a first linear-algebra or methods course; from Lecture 1 onward we build abstraction on top of that fluency rather than re-teaching it. The four topics below are the ones the early lectures lean on hardest. Complex arithmetic is used on the very first pages of Lecture 1 and becomes load-bearing from Week 7 (inner products, Pauli matrices, quantum phases); matrix algebra, determinants, and Gaussian elimination underpin everything from bases and rank to the characteristic polynomial of Week 5. None of this is new material—it is a fast recall pass, paired with the companion self-check (`exercises-0.pdf`) so you can find any rust before it costs you. Begin with its seven-check, 40–45 minute diagnostic route; use the sections here only to repair what it exposes, then complete the indicated recheck.

The course also assumes single-variable calculus (including integration by parts), elementary linear ODEs, sequences and series, and basic Fourier-series familiarity. The companion self-check routes those prerequisites separately; this refresher diagnoses algebraic readiness and is not a replacement analysis course.

Learning objectives

By the end of this lecture you will be able to:

- Add, multiply, conjugate, and divide complex numbers, and move freely between Cartesian and polar form.
- Multiply matrices, transpose them, and read off the trace, knowing which operations commute and which do not.
- Evaluate 2×2 and 3×3 determinants and use $\det A \neq 0$ as the test for invertibility.
- Row-reduce a linear system, describe its solution set, and find a basis for the solutions of $Ax = 0$.

1 Complex numbers

Standard quantum mechanics is naturally formulated over $\mathbb{C}$, so complex arithmetic is not optional background for its usual amplitude description. A *complex number* is written $z = a + bi$ with $a, b \in \mathbb{R}$ and $i^2 = -1$; its *real* and *imaginary* parts are $\Re z = a$ and $\Im z = b$. Addition is componentwise and multiplication follows from $i^2 = -1$ and the distributive law, e.g. $(2+i)(1+3i) = 2 + 6i + i + 3i^2 = -1 + 7i$.

1.1 Conjugate, modulus, division

The *conjugate* of $z = a + bi$ is $\overline{z} = a - bi$, and its *modulus* is

$$|z| = \sqrt{z\overline{z}} = \sqrt{a^2 + b^2}, \qquad \text{so} \qquad z\overline{z} = |z|^2.$$

Conjugation respects the arithmetic, $\overline{z + w} = \overline{z} + \overline{w}$ and $\overline{zw} = \overline{z}\,\overline{w}$, the modulus is multiplicative, $|zw| = |z|\,|w|$, and z is real exactly when $\overline{z} = z$. To divide by a *nonzero* complex number, multiply top and bottom by the conjugate of the denominator, turning it real:

$$\frac{3+2i}{1-i} = \frac{(3+2i)(1+i)}{(1-i)(1+i)} = \frac{3+3i+2i+2i^2}{1^2+1^2} = \frac{1+5i}{2} = \tfrac{1}{2} + \tfrac{5}{2}i.$$

1.2 Polar form and Euler's formula

Every nonzero complex number has a *polar form*

$$z = r\,e^{i\theta} = r(\cos\theta + i\sin\theta), \qquad r = |z|, \quad \theta = \arg z, \tag{0.1}$$

where $r > 0$ and the argument θ is determined modulo 2π. To get θ from $z = a + bi$: $\tan\theta = b/a$ fixes θ only up to π, so take $\theta = \arctan(b/a)$ when $a > 0$, add π when $a < 0$, and take $\theta = \pm\pi/2$ (sign of b) when $a = 0$. Sketching the point and reading its quadrant is the reliable check. The zero number is $0 = 0e^{i\theta}$ for every θ, but $\arg 0$ is undefined. The identity $e^{i\theta} = \cos\theta + i\sin\theta$ is *Euler's formula*. For nonzero factors, polar multiplication is almost free—moduli multiply and arguments add modulo 2π,

$$r_1e^{i\theta_1}\cdot r_2e^{i\theta_2} = r_1r_2\,e^{i(\theta_1+\theta_2)}, \qquad \text{so} \qquad \left(re^{i\theta}\right)^n = r^ne^{in\theta}, \qquad n\in\mathbb{Z},$$

(the second identity is *De Moivre's theorem*). Roots reverse this: a nonzero $z = re^{i\theta}$ has exactly n complex n-th roots,

$$z^{1/n} = r^{1/n}\,e^{i(\theta+2\pi k)/n}, \qquad k = 0, 1, \ldots, n-1,$$

obtained by taking the positive real n-th root of the modulus, dividing the argument by n, and then stepping round by $2\pi/n$. (Taking $z = 1$ recovers the roots of unity below.)

Example 0.1 (Powers via polar form). Write $-1+\sqrt{3}\,i$ in polar form and cube it. Here $r = \sqrt{1+3} = 2$ and the point lies in the second quadrant at $\theta = \frac{2\pi}{3}$, so $-1+\sqrt{3}\,i = 2\,e^{i2\pi/3}$. Then

$$\left(-1+\sqrt{3}\,i\right)^3 = 2^3e^{i\cdot 2\pi} = 8.$$

Cubing tripled the argument to a full turn and cubed the modulus—one line in place of a binomial expansion. ◁

Remark 0.2. The factor $e^{i\theta}$ is a complex number of modulus 1: a pure *phase*. Phases are the mathematical home of quantum interference, and the n-th *roots of unity*—the solutions of $z^n = 1$, namely $e^{2\pi ik/n}$ for $k = 0, \ldots, n-1$ and positive integers n—are the phases that recur whenever a finite cyclic symmetry appears. Standard quantum mechanics therefore uses complex Hilbert spaces naturally. A bare real vector space lacks multiplication by phases; a real formulation can recover the same structure only by adding a compatible real-linear operator J with $J^2 = -I$. We use the standard complex formulation from Week 7 onward.

2 Matrix algebra

Throughout, $\mathbb{F}$ denotes $\mathbb{R}$ or $\mathbb{C}$ — the field the entries are drawn from. A matrix in $M_{m\times n}(\mathbb{F})$ has m rows and n columns; matrices of the same shape add entrywise and scale entrywise. Two operations are worth recalling with care, because the course uses them constantly and because one of them does *not* behave like ordinary multiplication.

2.1 Multiplication

The product AB is defined only when the number of columns of A equals the number of rows of B; if A is $m\times n$ and B is $n\times p$ then AB is $m\times p$ with entries

$$(AB)_{ik} = \sum_{j=1}^{n} A_{ij}B_{jk}.$$

Multiplication is associative and distributes over addition. For an $m\times n$ matrix A, the correctly typed identity equations are $AI_n = A = I_mA$. (If A is $n\times n$, this becomes $AI_n = I_nA = A$.) But multiplication is *not commutative*:

$$\begin{pmatrix}1&1\\0&1\end{pmatrix}\begin{pmatrix}1&0\\1&1\end{pmatrix}=\begin{pmatrix}2&1\\1&1\end{pmatrix},\qquad \begin{pmatrix}1&0\\1&1\end{pmatrix}\begin{pmatrix}1&1\\0&1\end{pmatrix}=\begin{pmatrix}1&1\\1&2\end{pmatrix},$$

so in general $AB \neq BA$. Classical transformations can also fail to commute. In quantum mechanics, the non-commutative algebra of observables is one defining algebraic feature, used together with the theory's state and probability structure; non-commutativity alone does not characterize the theory.

2.2 Transpose and trace

The *transpose* A^{T} swaps rows and columns, $(A^{\mathsf{T}})_{ij} = A_{ji}$, and reverses products: $(AB)^{\mathsf{T}} = B^{\mathsf{T}}A^{\mathsf{T}}$. A matrix is *symmetric* if $A^{\mathsf{T}} = A$ and *antisymmetric* if $A^{\mathsf{T}} = -A$—two subspaces you will meet again in Lecture 1. For a square matrix A, the *trace* is the sum of the diagonal entries, $\operatorname{tr} A = \sum_i A_{ii}$; it is linear. More generally, if $A \in M_{m\times n}(\mathbb{F})$ and $B \in M_{n\times m}(\mathbb{F})$, both products are square and satisfy the cyclic identity

$$\operatorname{tr}(AB) = \operatorname{tr}(BA), \tag{0.2}$$

even though AB and BA can differ (and can even have different sizes). The *inverse* A^{-1}, when it exists, is the square matrix with $AA^{-1} = A^{-1}A = I$. For compatible invertible square matrices, inverses also reverse products, $(AB)^{-1} = B^{-1}A^{-1}$. *When* an inverse exists, and how to compute it, are the subjects of the next two sections.

Remark 0.3. Over $\mathbb{C}$ the operation that really matters is the *conjugate transpose* (or Hermitian adjoint) $A^\dagger = \overline{A}^{\mathsf{T}}$, which transposes and conjugates every entry. It is the matrix face of the adjoint $T^\dagger$ that organizes all of Weeks 9–10; for now simply note that the complex analogue of "symmetric" is $A^\dagger = A$ (Hermitian), not $A^{\mathsf{T}} = A$.

3 Determinants

To every *square* matrix the determinant assigns a scalar that detects invertibility. For 2×2 and 3×3 matrices the formulas are

$$\det\begin{pmatrix}a&b\\c&d\end{pmatrix} = ad - bc,\qquad \det\begin{pmatrix}a&b&c\\d&e&f\\g&h&i\end{pmatrix} = a\begin{vmatrix}e&f\\h&i\end{vmatrix} - b\begin{vmatrix}d&f\\g&i\end{vmatrix} + c\begin{vmatrix}d&e\\g&h\end{vmatrix},$$

the second being *cofactor (Laplace) expansion* along the top row. One may expand along any row or column, attaching the sign $(-1)^{i+j}$ to the entry in position (i, j); a good choice of row or column (one with zeros) saves work.

Example 0.4 (A 3×3 determinant). Expanding along the first row,

$$\det\begin{pmatrix}2&1&0\\1&3&1\\0&1&2\end{pmatrix} = 2(3\cdot 2 - 1\cdot 1) - 1(1\cdot 2 - 1\cdot 0) + 0 = 2\cdot 5 - 2 = 8 \neq 0,$$

so this matrix is invertible. ◁

3.1 The facts you will actually use

Three properties carry most of the weight. The determinant is *multiplicative*, $\det(AB) = \det A\,\det B$; it is unchanged by transpose, $\det A^{\mathsf{T}} = \det A$; and for a triangular matrix it is just the product of the diagonal entries. Row operations act predictably: swapping two rows flips the sign, scaling a row by c scales the determinant by c, and adding a multiple of one row to another leaves it unchanged (the basis of the next section). Above all,

$$\begin{aligned} A \text{ is invertible} &\iff \det A \neq 0 \\ &\iff \text{the columns of } A \text{ are linearly independent} \\ &\iff Ax = 0 \text{ only for } x = 0. \end{aligned}$$

For a 2×2 matrix the inverse is immediate from the determinant,

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix}^{-1} = \frac{1}{ad - bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix},$$

while in larger sizes row reduction (Section 4) is the practical route.

Remark 0.5 (Looking ahead to Week 5). The determinant is how eigenvalues will enter. The *characteristic polynomial* of a square matrix A is $\det(\lambda I - A)$—a monic polynomial in λ whose roots, the values of λ making $A - \lambda I$ singular, are exactly the eigenvalues. (Week 5 uses this monic form throughout.) Because you already know determinants, we can introduce eigenvalues this way in Week 5 without delay, rather than building up to them as Axler does.

4 Gaussian elimination

Gaussian elimination is the workhorse for solving linear systems, and from Week 2 it is also how we compute bases, rank, and dimension. The method applies three *elementary row operations*—swap two rows, scale a row by a nonzero constant, add a multiple of one row to another—to drive a system toward *reduced row-echelon form* (RREF): all zero rows lie below all nonzero rows; each leading nonzero entry of a nonzero row is a 1; each such leading 1 sits strictly to the right of the leading 1 above it; and each leading 1 is the only nonzero entry in its column.

4.1 Solving a system

Write the system as an augmented matrix $[A \mid b]$ and reduce. Check *consistency first*: a row $[\,0 \ \cdots \ 0 \mid c\,]$ with $c \neq 0$ signals an inconsistent system with no solution. Otherwise the system is consistent; a pivot in every variable column gives a unique solution, whereas at least one variable column without a pivot gives a free variable and hence infinitely many solutions.

Example 0.6 (A unique solution). For

$$\begin{aligned} x + 2y + z &= 4, \\ 2x - y + 3z &= 4, \\ -x + y + 2z &= 2, \end{aligned}$$

so, naming every operation—each arrow carries out its labels in the order written, the upper one first, which matters on the second and third arrows—

$$\left[\begin{array}{ccc|c} 1 & 2 & 1 & 4 \\ 2 & -1 & 3 & 4 \\ -1 & 1 & 2 & 2 \end{array}\right] \xrightarrow[R_3 \to R_3 + R_1]{R_2 \to R_2 - 2R_1} \left[\begin{array}{ccc|c} 1 & 2 & 1 & 4 \\ 0 & -5 & 1 & -4 \\ 0 & 3 & 3 & 6 \end{array}\right] \xrightarrow[R_2 \leftrightarrow R_3]{R_3 \to \frac{1}{3}R_3} \left[\begin{array}{ccc|c} 1 & 2 & 1 & 4 \\ 0 & 1 & 1 & 2 \\ 0 & -5 & 1 & -4 \end{array}\right]$$

$$\xrightarrow[R_3 \to \frac{1}{6}R_3]{R_3 \to R_3 + 5R_2} \left[\begin{array}{ccc|c} 1 & 2 & 1 & 4 \\ 0 & 1 & 1 & 2 \\ 0 & 0 & 1 & 1 \end{array}\right] \xrightarrow[R_1 \to R_1 - R_3]{R_2 \to R_2 - R_3} \left[\begin{array}{ccc|c} 1 & 2 & 0 & 3 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 \end{array}\right]$$

$$\xrightarrow{R_1 \to R_1 - 2R_2} \left[\begin{array}{ccc|c} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 \end{array}\right],$$

so $(x, y, z) = (1, 1, 1)$ is the unique solution; substituting it back satisfies all three original equations. This is the one place the refresher spells the elimination out in full — later examples compress the same work into a single $\xrightarrow{\text{RREF}}$ arrow. ◁

4.2 Homogeneous systems and the null space

The homogeneous system $Ax = 0$ is always consistent—$x = 0$ works—and it has *nontrivial* solutions precisely when there is a free variable. Its solution set is the *null space* of A, and it is a vector space (you will recognize it as one of the first examples in Lecture 1). Reading a basis off the RREF is a skill the course uses repeatedly.

Example 0.7 (A basis for the solution space). Take $A = \left(\begin{smallmatrix} 1 & 2 & -1 \\ 2 & 4 & -2 \end{smallmatrix}\right)$. The second row is twice the first, so the RREF is the single equation $x_1 + 2x_2 - x_3 = 0$. Treating $x_2 = s$ and $x_3 = t$ as free gives $x_1 = -2s + t$, hence

$$x = s\begin{pmatrix} -2 \\ 1 \\ 0 \end{pmatrix} + t\begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix},$$

so $\{(-2, 1, 0),\ (1, 0, 1)\}$ is a basis for the null space, which is therefore two-dimensional. Here A has *rank* 1 (the number of pivots) and *nullity* 2 (the dimension of the null space), and $1 + 2 = 3$ is the number of columns—a first glimpse of the rank–nullity theorem of Week 3. ◁

These two calculations combine. If $Ax = b$ is consistent and x_p is any one solution, then x solves $Ax = b$ exactly when $A(x - x_p) = 0$, so the full solution set is

$$x_p + \operatorname{null} A = \{x_p + v : Av = 0\}.$$

In practice: row-reduce $[A \mid b]$, set every free variable to 0 to read off a particular x_p, then set $b = 0$ and read off a basis of the null space as above. The answer is x_p plus an arbitrary combination of those basis vectors—the form the bank problems ask for.

Remark 0.8 (Inverses by row reduction). Row reduction also inverts a matrix: form the augmented block $[A \mid I]$ and reduce. If A is invertible the left half becomes I and the right half is then A^{-1}; if a zero row appears on the left, A is singular. This is the practical companion to the determinant test $\det A \neq 0$ of Section 3.

Summary of main concepts

The four skills above are the computational floor the course is built on. From $\mathbb{C}$ you need conjugate, modulus, nonzero division, and the polar form $re^{i\theta}$ for nonzero numbers, because the course's standard quantum models use complex amplitudes and phases. From matrix algebra you need multiplication (non-commutative!), transpose, and trace. Determinants give the one-number test $\det A \neq 0$ for invertibility and, in Week 5, the characteristic polynomial $\det(\lambda I - A)$. Gaussian elimination solves systems and—reading a basis off the RREF of $Ax = 0$—computes the null spaces, ranks, and dimensions that Lectures 1–4 turn into abstract structure. If the self-check exposes a gap, repair it here before Lecture 1.

- **Complex arithmetic**—$\overline{z}$, $|z| = \sqrt{z\overline{z}}$, division by a nonzero number using its conjugate, and nonzero $z = re^{i\theta}$ with arguments defined modulo 2π.
- **Matrix algebra**—$(AB)_{ik} = \sum_j A_{ij}B_{jk}$ (and $AB \neq BA$); $AI_n = I_m A = A$ for $A \in M_{m\times n}$; transpose A^{T}; trace $\operatorname{tr}(AB) = \operatorname{tr}(BA)$ for compatible rectangular factors.
- **Determinants**—2×2 and 3×3 by cofactors; $\det(AB) = \det A \det B$; $\det A \neq 0 \iff$ invertible $\iff$ columns independent.
- **Gaussian elimination**—complete RREF; inconsistency first, then pivots and free variables; the null space of $Ax = 0$ and a basis for it; rank as the pivot count.

The companion self-check is in `exercises-0.pdf`; take its seven-check diagnostic first, use the routing table to repair only the weak strand, and finish with the indicated recheck.